

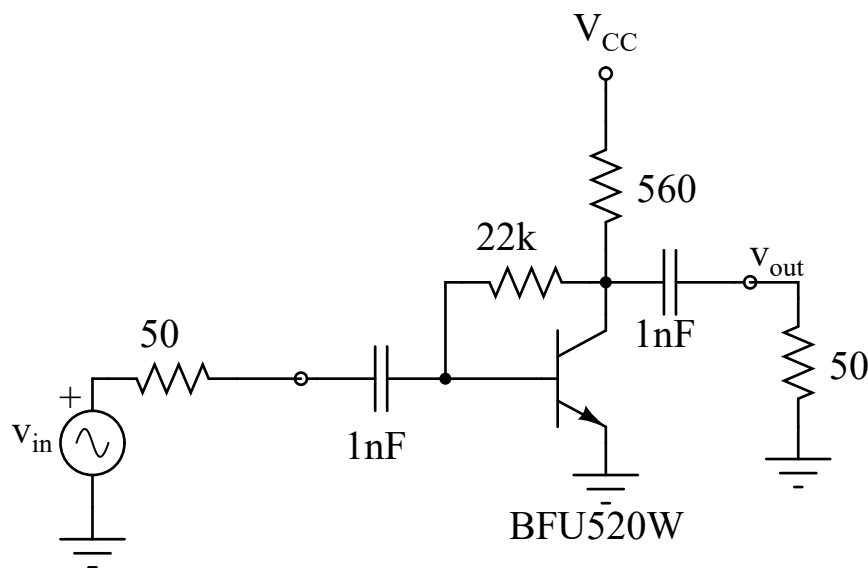
EE 412/512 Lab #2

Due December 17, 2025

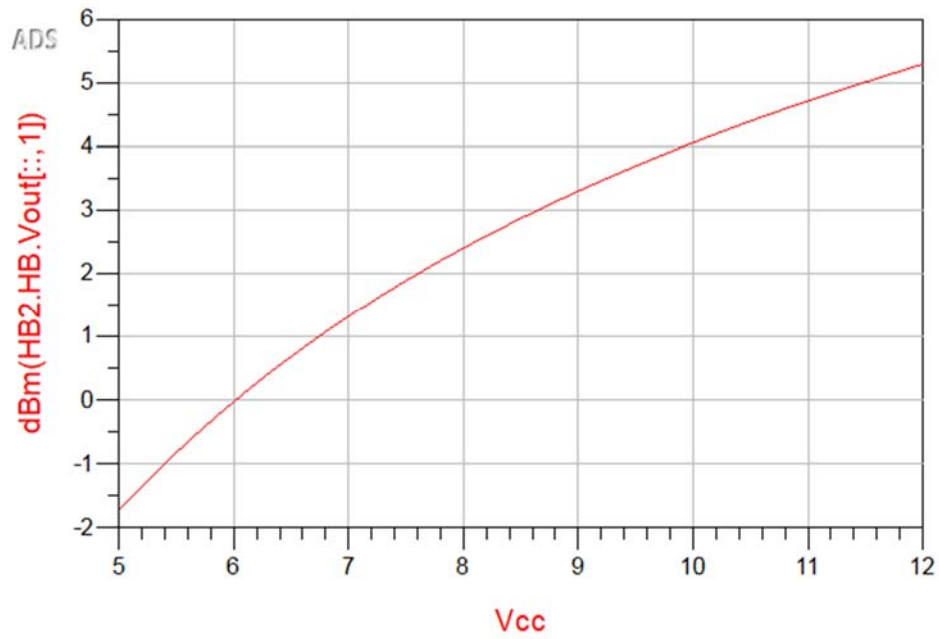
Write the following statement on the first line of your sheet:
 “I completed this lab assignment independently.”

Preliminary Work:

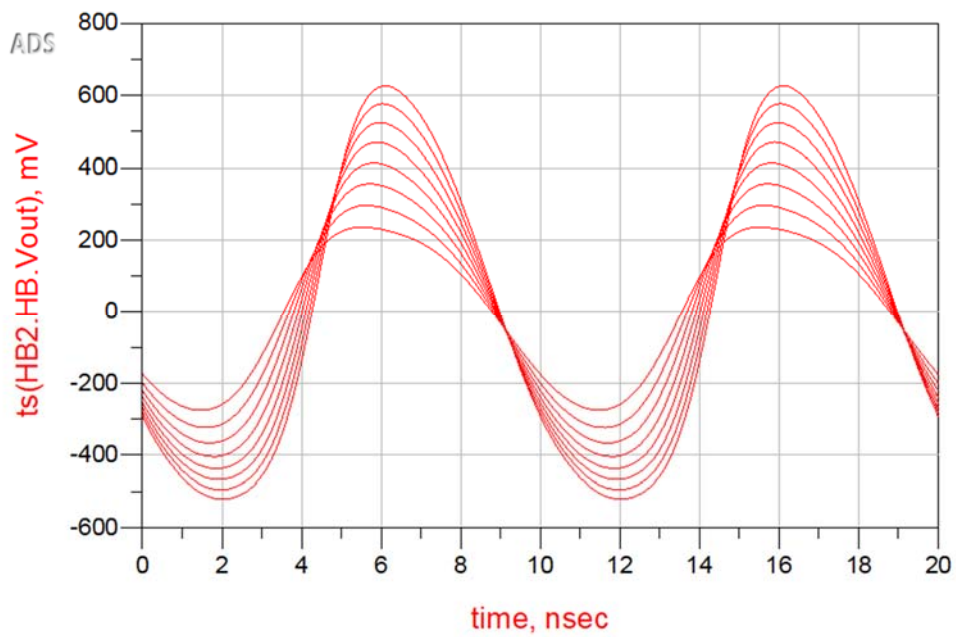
- 1) We have the following amplifier circuit. Choose an input frequency in the range 30MHz to 200MHz. Using the harmonic balance method in ADS, find the 1-dB compression point as a function of the supply voltage, V_{CC} , in the range 5V to 12V. Plot P_{1dB} as a function of the supply voltage V_{CC} .



Enter the schematic in ADS as shown. Set “ V_{CC} ”, “ pin ”, and “ fr ” as variables by placing them in the VAR block. Set their default values to 5V, -50, and the chosen frequency in Hz. Place P_1Tone block as the excitation source with its number set to 1. Set its Freq to fr . Set P to $polar(dBmtoW(pin),0)$. Connect a 50 ohm Term at the output with number set to 2. Place the HB block to perform a harmonic balance simulation. Set the Freq[1] to fr . Set Order[1]=5. Place the XDB block to find the 1dB gain compression point. Set the Freq, Input and Output frequencies to fr . Place a Parameter Sweep block to sweep the variable V_{CC} from 5V to 12V in 0.1V steps. Set its Simulation Instance to HB2 (the instance name of the XDB block). In the results window, plot HB2.HB.Vout and choose “Fundamental tone in dBm for all sweep values” as shown below. It should show the P_{1dB} as a function of the supply voltage V_{CC} .



Time-domain waveforms of the output voltage at the $P_{1\text{dB}}$ point for different V_{cc} values can be plotted by selecting the “time domain signal (one trace for each sweep)” option for the same quantity: HB2.HB.Vout. ---



- 2) Select a supply voltage in the range 5V to 12V. Set the second input frequency to be displaced by 1 MHz from the above-chosen input frequency. Find the output 3rd-order intercept point, OIP3, for both sidebands using ADS for different input power levels in the range of -50 dBm to -10 dBm.

Copy the first part's schematic to another schematic. Set V_{cc} variable default value to the chosen voltage. To excite with two sine waves, place P_nTone block with its number set to 1 as the excitation source rather than P_1Tone. Choose Freq[1] as fr. Add Freq[2] and choose Freq[2] value fr+1E6. Set the Freq[2] of the HB block to fr+1E6. Set P[1] of the P_nTone block to polar(dBmtoW(pin),0). Set P[2] to the same value. Remove XDB block.

Place the IP3out block in the schematic. Set ipo_1=ip3_out(vout;{1,0};{2,-1},50).

This definition finds the dBm value of vout when the fundamental frequency {1,0} (f_1) signal is equal to {2,-1} ($2f_1-f_2$) signal.

Set ipo_2= ip3_out(vout;{1,0};{-1,2},50). This finds the value of vout when f_1 signal equals ($2f_2-f_1$). Plot both quantities as a function of the input power pin as shown below:

